

Tashkent Ambulance Call Analysis and Prediction

1. Course & Team Details

- **Course Name:** AI
- **Team Name:** FARHOD

Team Members:

- **Leader:** XASAN OTAJONOV | ID: 202390249 | Group: E40A
- **Second Leader / Programmer:** NAZAROV NAZARBEEK | ID: 202390230 | Group: E40A
- **Programmer:** ABDULAZIZ AXMADJONOV | ID: 202390245 | Group: E40A
- **Programmer:** ABDULLAEV FARKHOD | ID: 202390250 | Group: E40B

2. Project Overview

- **Project Title:** Tashkent Ambulance Call Analysis and Prediction
- **Selected Field:** Option B — Data Analysis & Prediction (Polynomial Regression)

Dataset Details:

- **Dataset Title:** Monthly Ambulance Calls by Age Category and District in Tashkent City
- **Data Source:** Open Data Portal of Uzbekistan (data.gov.uz)
- **Dataset Size:** 510 rows × 17 columns
- **Selection Reason:** Fulfills the project requirement for localized Uzbekistan web data and supports multi-variable demographic analysis alongside non-linear regression modeling.

3. Objectives & Expected Outcomes

Objectives:

1. **Demographic Breakdown:** Identify which age demographics generate the highest volume of emergency medical requests.
2. **Geographic Distribution:** Pinpoint high-demand districts across Tashkent city for emergency dispatch optimization.
3. **Trend Forecasting:** Fit a 2nd-degree Polynomial Regression model to forecast monthly call trajectories and account for seasonal shifts.

Expected Outcomes:

1. **Predictive Model:** A working 2nd-degree Polynomial Regression model ($f(x) = ax^2 + bx + c$) trained to predict overall call volume in Tashkent.
2. **Visual Analytics:** Interactive line plots and scatter charts showing monthly call distributions and projected workload trends.

3. **Resource Insights:** Actionable data findings to help healthcare authorities allocate medical personnel and ambulance fleets effectively across Tashkent districts.

6. Project Timeline

Milestone / Week	Planned Activity	Deliverables
Week 1 (Oct 05 – Oct 7)	Topic selection & dataset retrieval	GitHub proposal creation
Week 2 (Oct 14 – Oct 15)	Data cleaning & feature setup	Prepared DataFrame in Pandas
Week 3 (Oct 16 – Oct 17)	Exploratory analysis & visualization	Aggregations & chart plots
Week 4 (Oct 18 – Oct 19)	Project Presentations	Report & presentation deck
Final Presentations	Submitting Final Report and Presentation Deck	October 21, 2026

7. Expected Outcomes & Skills

- Expertise in handling real-world structured datasets using Python ([pandas](#), [numpy](#), [matplotlib](#)).
- Practical understanding of non-linear machine learning models and coefficient interpretation ($f(x) = ax^2 + bx + c$).

8. References

1. Open Data Portal of Uzbekistan ([data.gov.uz](#)).
2. NumPy & Pandas Documentation ([numpy.org](#), [pandas.pydata.org](#)).
3. GitHub Repository:
<https://github.com/Nonamechick/Tashkent-Ambulance-Call-Analysis-and-Prediction>